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(54) **GATA3 INHIBITORS FOR TREATING INSULIN RESISTANCE**

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(58) **Field of Classification Search**

CPC **C12N 15/1138**; **A61K 9/0019**; **A61K 9/0053**; **A61P 3/10**

See application file for complete search history.

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(57) **ABSTRACT**

Methods and compositions using dnayzme-based GATA3 inhibitors for the treatment of insulin resistance and Type II diabetes, for the stimulation of adipogenesis in vivo, and for the promotion of fat redistribution are disclosed.

10 Claims, 4 Drawing Sheets

Specification includes a Sequence Listing.

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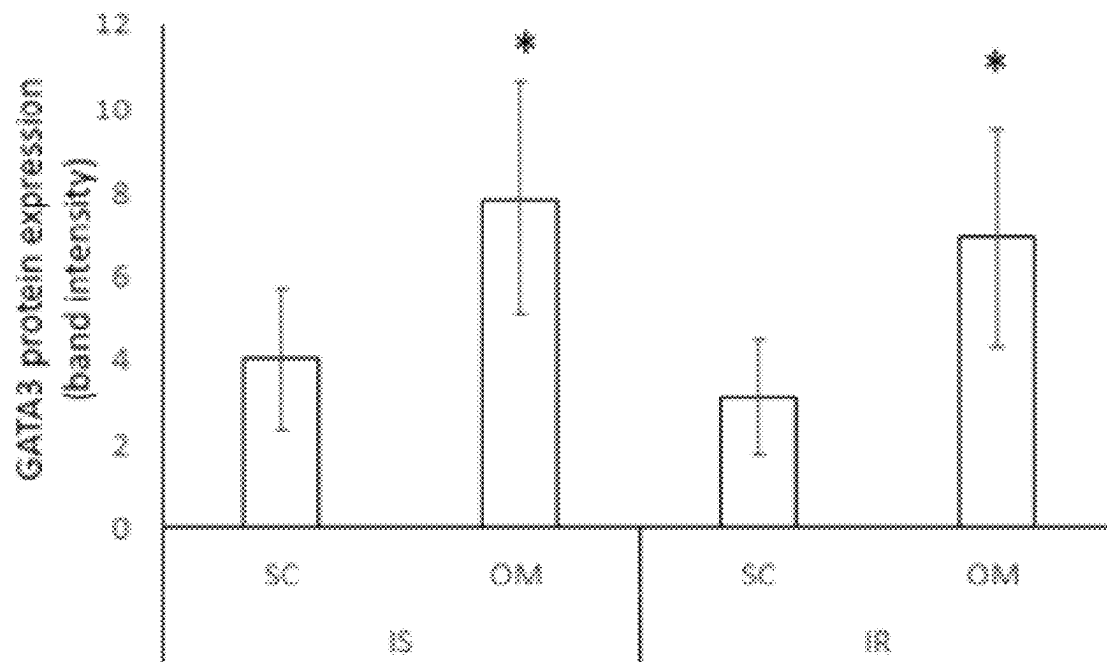


Figure 1

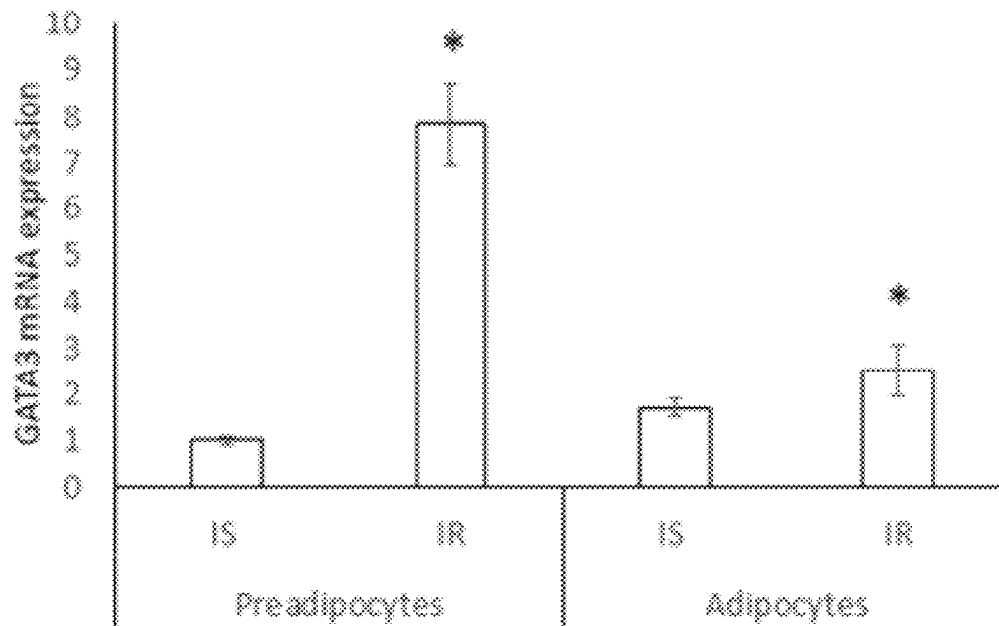


Figure 2

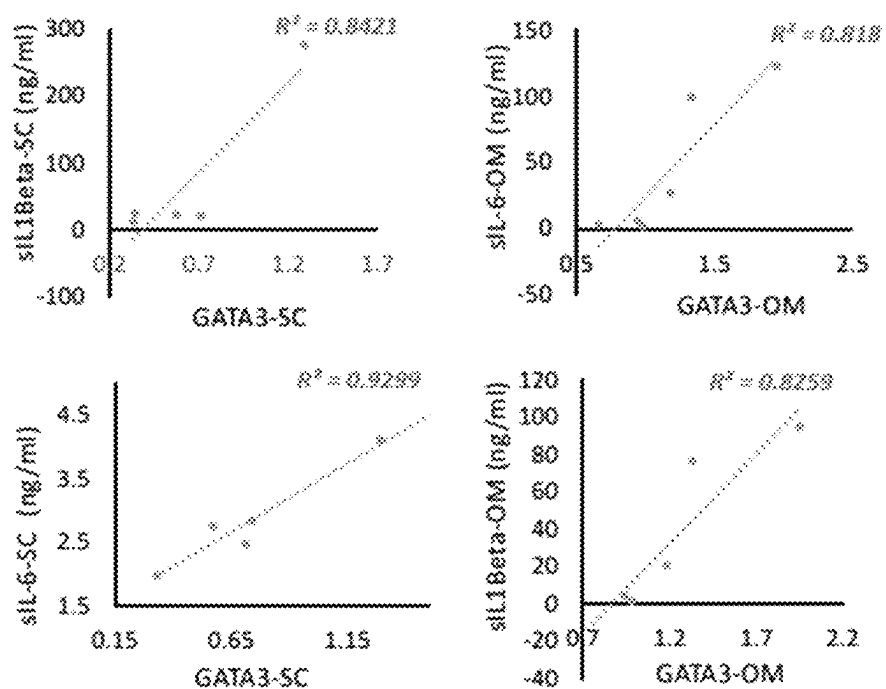


Figure 3

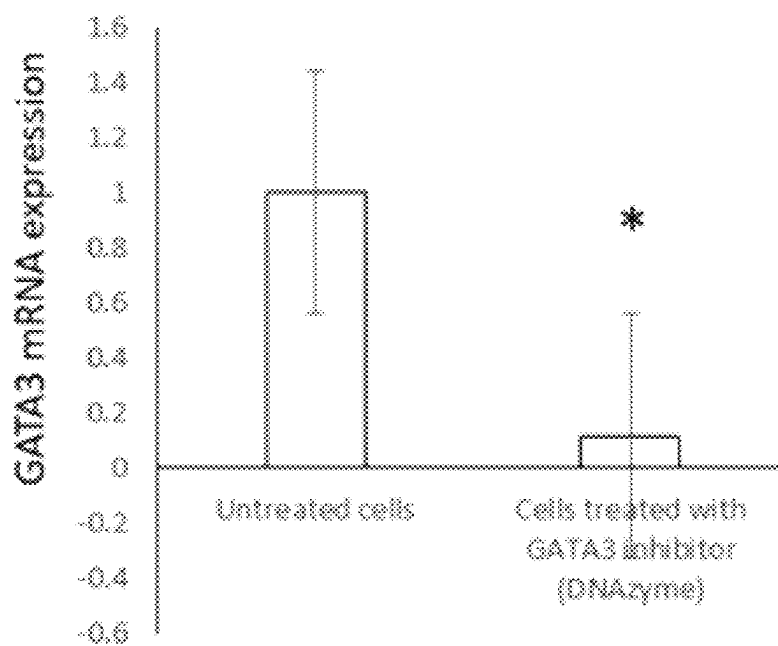


Figure 4

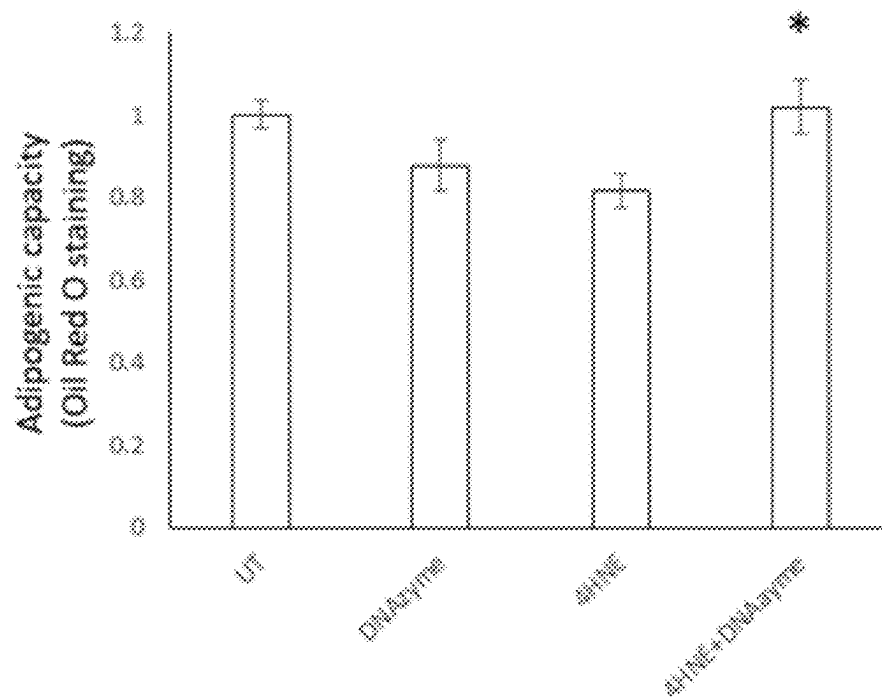


Figure 5

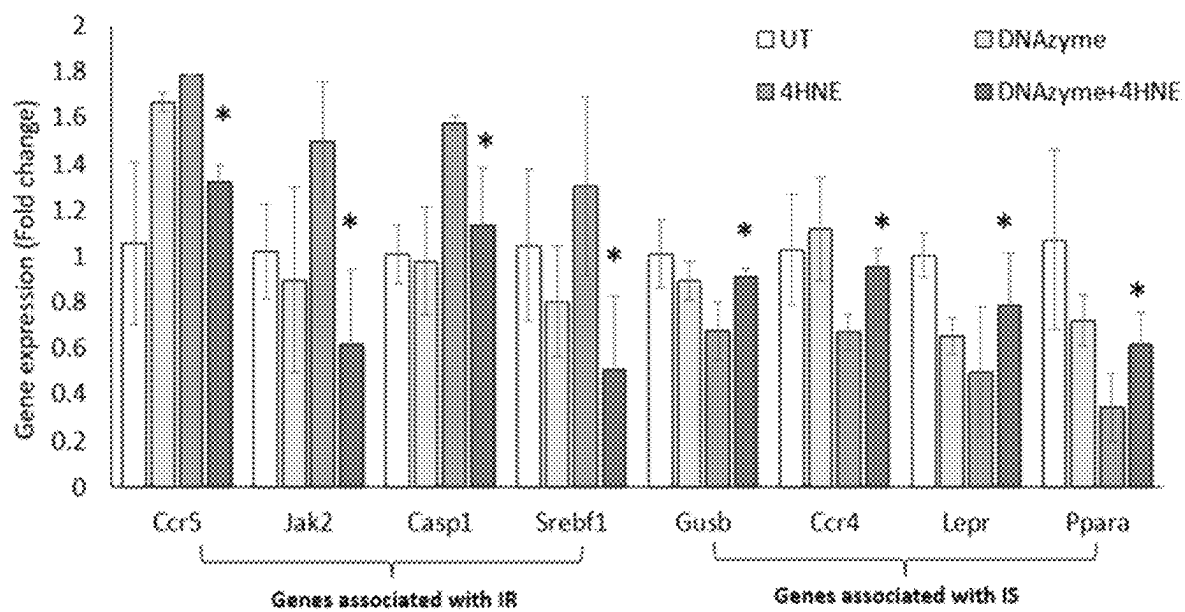


Figure 6

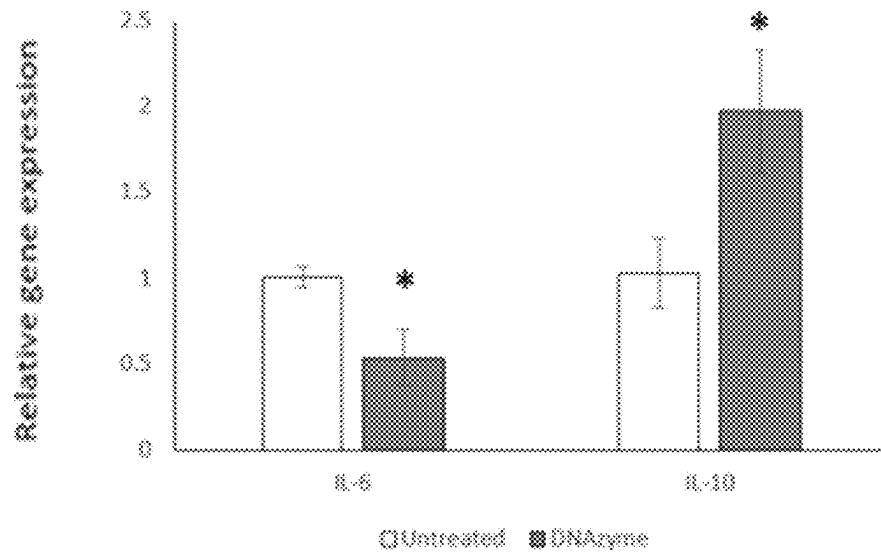


Figure 7

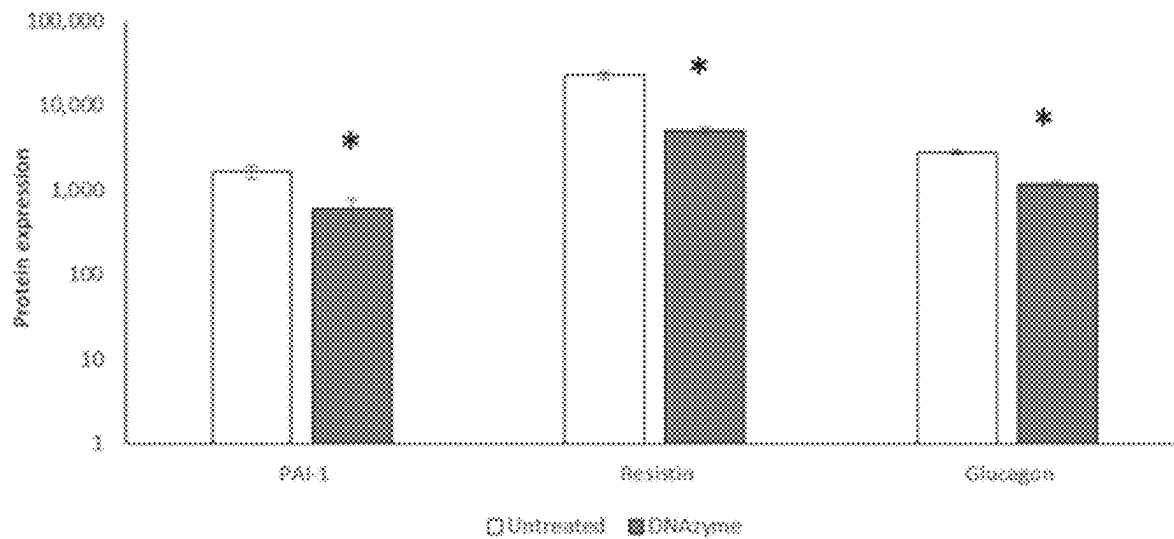


Figure 8

1

**GATA3 INHIBITORS FOR TREATING
INSULIN RESISTANCE****CROSS-REFERENCE TO RELATED
APPLICATION(S)**

This application is a divisional application of U.S. patent application Ser. No. 16/909,755 filed on Jun. 23, 2020, the entire content of which is incorporated by reference.

**REFERENCE TO ELECTRONIC SEQUENCE
LISTING**

The application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Sep. 5, 2023, is named "432743.10347ST26.xml" and is 203,704 bytes in size. The sequence listing contained in this XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

FIELD OF INVENTION

Provided herein are methods relating to the use of GATA3 inhibitors for the treatment of insulin resistance and related disorders and diseases, including Type II diabetes, as well as to methods for using GATA3 inhibitors to stimulate adipogenesis in vivo and to stimulate fat redistribution from visceral deposits to subcutaneous deposits in desirable locations in the body.

BACKGROUND**GATA3**

GATA3 is part of the GATA family of transcription factors, which play roles in controlling expression in a wide variety of genes. The GATA3 gene is comprised of six exons encoding a transcription factor with two transactivation domains and two zinc finger domains (see, e.g., Joulin et al., *EMBO J.*, 10(7):1809-16 (1991), Yamashita et al., *JBC*, 279(26):26983-(2004), and Van Esch et al., *Nature*, 406(6794):419-22 (2000), that binds to the consensus sequence 5'-WGATAG-3', and is expressed in T cells, mast cells, basophils, epithelial cells and fat cells, including white adipocyte precursors. The role of GATA3 in Treg cellular plasticity and conversion into Th1-Treg or APC-T-reg has been extensively characterized (Xu et al., *Front. Immunol.*, 9:45 (2018)).

GATA3 is necessary for the development of MHC II-restricted CD4 T cells from CD4+ cells. It is also associated with normal development of various tissues including the skin, fat cells, the thymus, and the nervous system. GATA3 is expressed in a range of cells found in adipose tissues, including preadipocytes, mature adipocytes and various inflammatory cells.

GATA3's role in the conversion of CD4+ cells into TH2 cells has been utilized in the design of asthma therapies, based on the inhibition of GATA3 to prevent inflammatory diseases. A variety of GATA3 inhibitors have been used as therapeutic compositions.

Hong et al., reported that ethanol extracts of *Holotrichia diomphalia* reduced GATA3 expression in lungs in a mouse model of allergic asthma. Ghonim et al. reported that administration of the drug olaparib reduced GATA3 expression in a mouse asthma model (Ghonim et al., *Clin. Sci.*, 29(11): 951-62 (2015)). The naturally occurring small molecule bavachinin has also been found to reduce GATA3 expression

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in the mouse asthma model (Chen et al., *Cellular and Molecular Immunology*, 10:497-505 (2013)). Sel et al (2008) *J. Allergy Clin. Immunol.*, 121(4):910-916, describes the use of the dnzyme-based GATA3 inhibitor ggd21 to prevent airway inflammation and mucus production and inhibit development of airway hyperresponsiveness to methacholine in models of acute allergic airway inflammation.

Dnzyme-based GATA3 inhibitors are also used to target other TH2-based inflammatory-related diseases. SB011, from Sterna Biologicals, is a DNzyme that targets GATA3 and cleaves its messenger RNA. It is currently in Phase II clinical trials as both part of an inhalant therapy for the treatment of asthma and chronic obstructive pulmonary disease (COPD). It is also currently in Phase II studies as part of a composition for topical administration for the treatment of atopic dermatitis. Patents U.S. Pat. Nos. 9,249, 413 and 9,404,110 are directed to pharmaceutical compositions for topical application comprising DNzymes with GATA3 inhibitory activity with defined sequences. A second dnzyme-based pharmaceutical with GATA3 inhibitory activity by Sterna, SB012, is in Phase II trials as a treatment for active ulcerative colitis.

In addition to being involved in TH2-based inflammatory responses, GATA3 has been identified as a factor associated with obesity (Tong et al., *Science*, 290(5489):134-8 (2000)). GATA3 forms protein complexes with CCAAT/enhancer binding proteins alpha (C/EBPa) and beta (C/EB113), members of a family of transcription factors that are integral to adipogenesis. GATA3 expression is upregulated in insulin resistant obese individuals in comparison to insulin sensitive of similar weight (Almuraikhy et al., *Diabetologia*, 59:2406-16 (2016)). There is a clear need for improved strategies to treat for the treatment of obesity-related diseases and conditions, and, GATA3 is an attractive therapeutic target for the development of new therapies to reverse insulin resistance and lower the risk of Type II diabetes.

SUMMARY OF THE INVENTION

Provided herein are methods for treatment of insulin resistance and related diseases and disorders thereof by inhibiting GATA3 expression. In some embodiments, GATA3 expression inhibited by administration of an effective amount of a GATA3 inhibitor. GATA3 inhibitors of the disclosure include small molecules, antibodies, dnzymes or siRNAs. One or more GATA3 inhibitors can be used in therapeutic compounds of the disclosure. In certain embodiments, the GATA3 inhibitor is a dnzyme, e.g., hgd40. In some embodiments is a human patient.

Also provided herein are methods of stimulating adipogenesis in an individual by inhibiting GATA3 expression. In some embodiments GATA3 expression is inhibited by administration of an effective amount of a GATA3 inhibitor. GATA3 inhibitors for stimulating adipogenesis include small molecules, antibodies, dnzymes siRNAs. Of particular interest is the use of GATA3-inhibiting dnzymes to stimulate adipogenesis in individuals. In some embodiments, the dnzyme is hgd40.

Also provided herein are methods of stimulating fat redistribution from one or more visceral adipose tissue deposits in a human patient to a desired subcutaneous target site in said patient by administration of an effective amount of one or more GATA3 inhibitors to a site at or adjacent to the desired target site for fat redistribution. In some embodiments, the fat redistribution is mediated by the administration of one or more GATA3 inhibitors in, at or adjacent to a tissue where increased adipogenesis is desired. GATA3

inhibitors for stimulating fat redistribution include small molecules, antibodies, dnayzmes siRNAs. Of particular interest is the use of GATA3-inhibiting dnayzmes to stimulate adipogenesis in individuals. In some embodiments the dnayzyme is hgd40.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 shows GATA3 expression in subcutaneous (SC) and omental (OM) tissues by Western blotting.

FIG. 2 shows GATA3 expression in expanded preadipocytes/adipocytes from SC tissues from insulin sensitive (IS) and insulin resistant (IR) individuals by RT-PCR in human adipose tissue from insulin sensitive (IS) and insulin resistant (IR) individuals.

FIG. 3 shows the correlation between GATA3 mRNA of secreted inflammatory cytokines IL1f3, IL-6.

FIG. 4 shows the effect of a GATA3 inhibitor on GATA3 mRNA expression levels.

FIG. 5 shows the effect of 4HN treatment on preadipocytes differentiation.

FIG. 6 shows compares relative mRNA expression levels of selected insulin signaling genes in mouse 3T3L1 cells in vitro.

FIG. 7 shows the effect of a GATA3 inhibitor on inflammatory biomarkers. *P<0.05, independent sample t-test.

FIG. 8 shows the effect of a GATA3 inhibitor on insulin resistance-related biomarkers. *P<0.05, independent sample t-test.

DETAILED DESCRIPTION

Definitions

When referring to the compounds and methods provided herein, the following terms have the meanings as defined herein. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of ordinary skill in the art. In the event that multiple definitions for a term are used herein, definitions in this section will prevail, unless otherwise stated. The techniques and procedures described or referenced herein are generally understood and commonly employed using conventional methodologies described in Green and Sambrook, *Molecular Cloning: A Laboratory Manual* 4th ed. (2012), Cold Spring Harbor Laboratory Press, Cold Spring harbor, NY; Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley & Sons; and Rosenberg et al. As appropriate, procedures involving the use of commercially available kits and reagents are generally carried out in accordance with manufacturer-defined protocols and conditions unless otherwise noted.

As used herein, the singular forms “a,” “an,” and “the” include the plural references unless the context clearly indicates otherwise.

The term “about” indicates and encompasses an indicated value and a range above and below that value. In certain embodiments, the term “about” indicates the designated value $\pm 10\%$, $\pm 5\%$, or $\pm 1\%$. In certain embodiments, the term “about” indicates the designated value $\pm$ one standard deviation of that value. For logarithmic scale, “about” indicates a designated value ± 0.1 or ± 0.2 log units.

“Treating” or “treatment” of any disease or disorder refers, in certain embodiments, to ameliorating a disease or disorder, either physically (e.g., stabilization of a discernable symptom) or physiologically (e.g., stabilization of a physical parameter) or both. In yet another embodiment, “treating” or “treatment” of a disease refer to executing a

protocol, which may include administering one or more therapeutic agents to an individual (human or otherwise), in an effort to obtain beneficial or desired results in an individual, including clinical and cosmetic results. Beneficial or desired clinical results include, but are not limited to, alleviation or amelioration of one or more symptoms, diminishment of extent of disease(s), stabilized (i.e., not worsening) state of disease, preventing spread or increase in severity of disease, delay or slowing of disease progression, amelioration or palliation of the disease state, and remission (whether partial or total). “Treatment” can also mean prolonging survival as compared to expected survival of an individual not receiving treatment. Further, “treating” and “treatment” may occur by administration of one dose of a therapeutic agent or therapeutic agents, may occur by administration of one dose of a therapeutic agent or therapeutic agents, or may occur upon administration of a series of doses of a therapeutic agent or therapeutic agents. “Treating” or “treatment” does not require complete alleviation of signs or symptoms, and does not require a cure. “Treatment” can also refer to clinical intervention, such as administering one or more therapeutic agents to an individual, designed to alter the natural course of the individual being treated (i.e., to alter the course of the individual that would occur in absence of the clinical intervention).

As used herein, “cosmetic” refers to a beautifying substance or preparation, which preserves, restores, bestows, simulates, or enhances the appearance of bodily beauty or appears to enhance the beauty or youthfulness, specifically as it relates to the appearance of tissue or skin.

As used herein, “cosmetically effective amount” means an amount of a physiologically active compound or composition sufficient for treating one or more conditions, but low enough to avoid serious side effects. The cosmetically effective amount of the compound or composition will vary with the particular condition being treated, the age and physical condition of the end user, the severity of the condition being treated/prevented, the duration of the treatment, the nature of other treatments, the specific compound or product/composition employed, the particular cosmetically-acceptable carrier utilized, and like factors.

As used herein, “cosmetically acceptable” means that the ingredients the term describes are suitable for use in contact with tissues (e.g., the skin) without undue toxicity, incompatibility, instability, irritation, allergic response, or the like.

As used herein, a “cosmetically acceptable active agent” is a compound (synthetic or natural) that has a cosmetic or therapeutic effect on the skin.

The term “apply” or “application” as used in reference to a composition, means to apply or spread the compositions of the present invention onto a substrate such as the human skin surface or epidermis.

As used herein, “topical application” means directly laying on or spreading on outer skin or the scalp, e.g., by use of the hands or an applicator such as a wipe, roller, or spray.

The term “dermatologically acceptable” as used herein means that the compositions or components described are suitable for use in contact with human skin tissue without undue toxicity, incompatibility, instability, allergic response, and the like.

The term “safe and effective amount” as used herein means an amount of a compound or composition sufficient to significantly induce a positive benefit.

As used herein, the terms “prophylactic agent” and “prophylactic agents” refer to any agent(s) which can be used in the prevention of a condition, disease, or disorder or one or more symptoms thereof. In certain embodiments, the term

“prophylactic agent” includes a compound provided herein. In certain other embodiments, the term “prophylactic agent” does not refer to a compound provided herein. In certain embodiments, a prophylactic agent can be an agent which is known to be useful for, or has been or is currently being used to prevent or impede the onset, development, progression and/or severity of a condition, disease, or disorder associated with insulin resistance, or which is known to be useful for, or has been or is currently used to prevent or impede the onset of side effects or reduce the severity of side effects.

An “effective amount” of an agent disclosed herein is an amount sufficient to carry out a specifically stated purpose. An “effective amount” may be determined empirically and in a routine manner, in relation to the stated purpose. An “effective amount” or an amount “sufficient” of an agent is that amount adequate to produce a desired biological effect, such as a beneficial result, including a beneficial clinical result. In some embodiments, the term “effective amount” refers to an amount effective to “treat” a disease or disorder in an individual (e.g., a mammal such as a human).

As used herein, the term “therapeutically effective amount” or “effective amount” refers to amount of a protein, nucleic acid, oligonucleotide, or composition that, when administered to a subject is effective to treat a disease or disorder. In some embodiments, a therapeutically effective amount of a composition that, when administered to a subject is effective to prevent or ameliorate a disease or progression of the disease, or result in amelioration of symptoms. The term “therapeutic agent” can refer to one or more GATA3 inhibitors, or combinations or compositions thereof.

As used herein, the phrase “prophylactically effective amount” refers to the amount of a therapy (e.g., prophylactic agent) which is sufficient to result in the prevention or reduction of the development, recurrence or onset of one or more symptoms associated with a condition, disease, or disorder, or reduce or prevent side effects, or to enhance or improve the prophylactic effect(s) of another therapy (e.g., another prophylactic agent).

“Administration” and variants thereof (in some embodiments, “administering” a compound) in reference to a compound, e.g., a GATA3 inhibitor, means introducing the compound or a prodrug of the compound, or a population of cells treated with said compound, e.g., a GATA3 inhibitor, into the system of the animal. When a compound of the disclosure or prodrug thereof is provided in combination with one or more other active agents, “administration” and its variants are each understood to include concurrent and sequential introduction of the compound or prodrug thereof and other agents or elements.

As used herein, “adipogenesis” is the formation of fat cells (adipocytes) from stem cells in an individual.

As used herein, “adipogenic capacity” measures the potential of preadipocytes found in adipose tissue to differentiate into adipocytes.

DNAzymes, also known as DNA enzymes, or catalytic DNA, refer to DNA oligonucleotides that are capable of performing a specific chemical reaction, often but not always catalytic.

“Insulin sensitivity” describes how sensitive the body is to the effects of insulin. Someone said to be insulin sensitive will require smaller amounts of insulin to lower blood glucose levels than someone who has low sensitivity.

“Insulin resistance” is an impaired, or reduced, response of the body to insulin, resulting in elevated levels of glucose in the blood (a key component of type 2 diabetes and metabolic syndrome).

The terms “pharmaceutical formulation” and “pharmaceutical composition” refer to preparations that are in such form as to permit the biological activity of the active ingredient to be effective, and that contain no additional components that are unacceptably toxic to an individual to which the formulation or composition would be administered. Such formulations or compositions may be sterile.

“Excipients” as used herein include pharmaceutically acceptable excipients, carriers, vehicles or stabilizers that are nontoxic to the cell or mammal exposed thereto at the dosages and concentrations employed. Often the physiologically acceptable excipient is an aqueous pH buffered solution.

Reference to a compound as described in a pharmaceutical composition, or to a compound as described in a claim to a pharmaceutical composition, refers to the compound described by the formula recited in the pharmaceutical composition, without the other elements of the pharmaceutical composition, that is, without carriers, excipients, etc.

As used herein, the terms individual, and patient are used interchangeably. The term “individual” refers to an animal, such as a mammal including a non-primate (e.g., a cow, pig, horse, cat, dog, rat or mouse) and a primate (e.g., a monkey such as a cynomolgous monkey, a chimpanzee and a human), and in certain embodiments, a human.

A “small molecule” as used herein refers to a compound of 1000 daltons or less in molecular weight.

GATA3 Inhibitors

GATA3 inhibitors that can be used in the embodiments disclosed herein include, but are not limited to, small molecules, antibodies, and antisense molecules including siRNA, ribozymes and deoxyribozymes (dnazymes).

Small molecule-based GATA3 inhibitors that may be used in embodiments describe herein include synthetic molecules as well as molecules isolated from naturally occurring compounds. For example, molecules displaying GATA3 inhibitory activity, e.g., such as *Holotrichia diomphalia* can be used. Pyrrothiogatain ((2,5-Dimethyl-1H-pyrrol-1-yl) thiophene-2-carboxylic acid) has also been shown to act as a GATA3 inhibitor (Nomura et al., *Scientific Reports*, 9: 17335 (2019)) and may be used in embodiments described herein. Corticosteroids may also be used to inhibit GATA3 in the compositions and methods described herein.

Anti-GATA3 antibodies, as exemplified but not limited to those as described in U.S. Pat. No. 9,429,577, are also used in some compositions and methods described herein. Anti-GATA3 polyclonal and anti-GATA3 monoclonal antibodies can be used in the compositions and methods described herein.

siRNA-based nucleic acids with GATA3 inhibitory activity as exemplified, by, but not limited to WO2019217423 may be used in compositions and methods as described herein.

DNAzymes (also known as deoxyribozymes, DNA enzymes, catalytic DNA, or DZ) can also be used as GATA3 inhibitors. For example, dnazymes with GATA3 inhibitory activity, as exemplified, but not limited to those as disclosed in U.S. Pat. Nos. 8,119,789, 9,249,413, 9,404,110, 9,428,576, 9,417,243, U.S. Ser. No. 10/273,526, U.S. Ser. No. 10/487,325, and US20200131267 may be used in compositions and methods as described herein. In some embodiments, compositions comprising more than one dnzyme with GATA3 inhibitory activity is used. For example, the dnzyme hgd40 (5'-GTGGATGGAGGCTAGCTA-CAACGAGTCTTGAG-3') (SEQ ID NO:1) can be used in methods and compositions described herein. Table 1 below

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provides other dnazymes that can be used in the methods and compositions described herein.

TABLE 1

Dnazyme sequence	Sequence identifier
GTGGATGGAGGCTAGCTACAACGAGTCTTGGAG	1
TCACGGCAAGGCTAGCTACAACGAGAACTGGGT	2
TCGGTCAGAggctagctacaacgaTGCGTTGCT	3
GGCGTACGAggctagctacaacgaCTGCTCGGT	4
GGCGGCGTAggctagctacaacgaGACCTGCTC	5
CTCGGGTCaggctagctacaacgaCTGGGTAGC	6
TCCTCTGCaggctagctacaacgaCGGGTCTCT	7
ACTCTGCAAggctagctacaacgaTCTGCGAGC	8
GGCGGACGaggctagctacaacgaTCTGCAATT	9
AAGGGGCGAggctagctacaacgaGACTCTGCA	10
AAAACGGGAggctagctacaacgaCAGGTTGTA	11
AGAATAAAAggctagctacaacgaGGGACCAGG	12
ATGGCAGAAggctagctacaacgaAAAACGGGA	13
ATCCAAAAAggctagctacaacgaTGGGTATGG	14
AGGGGAAGaggctagctacaacgaAAAAATCCA	15
TTTTAAAAAggctagctacaacgaTATCTTGGA	16
GTGGGGGAggctagctacaacgaGGGAAGGCT	17
GTGAATGAggctagctacaacgaTTGCTTTCG	18
GTCGTTGAaggctagctacaacgaGATTGCTT	19
GGCCCCGAaggctagctacaacgaCCGCGCGCG	20
TCACCTCCaggctagctacaacgaGGCCTCGGC	21
CCGCCGTCaggctagctacaacgaCTCCATGGC	22
GGTGGCTCaggctagctacaacgaCCAGCGCGG	23
CGTTGAGCaggctagctacaacgaGGCGGGTG	24
CCGCGTCCaggctagctacaacgaGTAGGAGTG	25
CAGCGGGTAggctagctacaacgaTGCGCCGCG	26
GCACATCCaggctagctacaacgaTCCTCCGG	27
AAAAGCACaggctagctacaacgaCCACCTCCT	28
TAAAAAGCaggctagctacaacgaATCCACCTC	29
GACCGTCGaggctagctacaacgaGTTAAAAAG	30
TTGCCTTGaggctagctacaacgaCGTCGATGT	31
AGGGCGGGaggctagctacaacgaGTGGTTGCC	32
TGGCCCTGaggctagctacaacgaCGAGTTTCC	33
ACCTCTGCaggctagctacaacgaCGTGCCCT	34
CGGAGGGTAggctagctacaacgaCTCTGCACC	35
GGCGGCACaggctagctacaacgaCTGGCTCCC	36
CGGGCGGCaggctagctacaacgaACCTGGCTC	37

8

TABLE 1-continued

Dnazyme sequence	Sequence identifier
AGGGATCCAggctagctacaacgaGAAGCAGAG	38
GGGTAGGGAggctagctacaacgaCCATGAAGC	39
GGGCTGAGAggctagctacaacgaTCCAGGGGG	40
GTGGATGGAggctagctacaacgaGTCTTGGAG	41
CGTGGTGGAggctagctacaacgaGGACGTCTT	42
GGGGGTAGAggctagctacaacgaGGAGAGGGG	43
GGAGGAGGAggctagctacaacgaGAGGCCGGG	44
GGCCCCGAggctagctacaacgaAAGGAGGAG	45
CCGGGGAGAggctagctacaacgaGTCTTCGG	46
GGACAGCGAggctagctacaacgaGGGTCCGGG	47
TGGGGTGGAggctagctacaacgaAGCGATGGG	48
CTTGAGGCaggctagctacaacgaTCTTTCTCG	49
CACCTGGTAggctagctacaacgaTTGAGGCAC	50
GCAGGGGCaggctagctacaacgaCTGGTACTT	51
CCAGCTTCaggctagctacaacgaGCTGTCCGG	52
GTGGGACGAggctagctacaacgaTCCAGCTTC	53
GGAGTGGGAggctagctacaacgaGACTCCAGC	54
ATGCTGCCaggctagctacaacgaGGGAGTGGG	55
GGGCGGTCaggctagctacaacgaGCTGCCACG	56
GAGGCTCCaggctagctacaacgaCCAGGGCGG	57
GTGGGTCGaggctagctacaacgaGAGGAGGCT	58
AGGTGGTGaggctagctacaacgaGGGTGGTG	59
TGGTTCTAggctagctacaacgaGCCCTCGTC	60
GGGCTCTGaggctagctacaacgaGCCTGGCTT	61
GGGACCCCaggctagctacaacgaCGGAGCCCG	62
GGTGGGGAggctagctacaacgaCCCACCGGA	63
GGGGGGGAggctagctacaacgaCCGAGGGCC	64
GGGCTGGGAggctagctacaacgaGGGCAGGGA	65
CGTCGAGGaggctagctacaacgaCCGCCCTC	66
GGGCTGGCaggctagctacaacgaCTTCCCGTA	67
CGATGCCCaggctagctacaacgaCCGGGGCGG	68
GCTCCACGaggctagctacaacgaGCCCATCCG	69
CCGGCTCCaggctagctacaacgaGATGCCCAT	70
TCTCCGCAAggctagctacaacgaCCGGCTCCA	71
CCGTGAGCaggctagctacaacgaGTCTCCGCA	72
TCCCCGGCaggctagctacaacgaCGGCTCGGT	73
ACTCGGGCaggctagctacaacgaGTAGGGGGG	74
GGAGCTGTAggctagctacaacgaTCGGGCACG	75

9

TABLE 1-continued

Dnzyme sequence	Sequence identifier
GGACTTGCAGgctagctacaacgaCCGAAGCCG	76
GGGCCTGGAggctagctacaacgaTTGCATCCG	77
TGTGCTGGAggctagctacaacgaCGGCCTTG	78
GTTACACAggctagctacaacgaTCCCTGCCT	79
CAGTTCACAggctagctacaacgaACTCCCTGC	80
CACAGTTCAggctagctacaacgaACACTCCCT	81
GTTGCCCCAggctagctacaacgaAGTTCACAC	82
TCGCCGCCAggctagctacaacgaAGTGGGGTC	83
CCCGTGCCAggctagctacaacgaTCGCCGCC	84
GGCGTTGCAGgctagctacaacgaAGGTAGTGT	85
TGGCTTCTAggctagctacaacgaGCCCTCGTC	86
CCCCCGGAggctagctacaacgaGCTCGTCCG	87
GTAGGGAGAggctagctacaacgaCCGAGGCTG	88
GGGCGGGCAGgctagctacaacgaCAAGCGCC	89
CGGGAAGGAggctagctacaacgaTCGCCGCCG	90
TAGTCCTCAGgctagctacaacgaGCGGCCCG	91
TCCCCGACAggctagctacaacgaTCCAGTCC	92
TTTCCCCGAggctagctacaacgaACCTCCAGT	93
TGAGCGCGAggctagctacaacgaCCTCAGTTT-	94
GGACCACAAGgctagctacaacgaAGGTGGTTG	95
CTTGGACCAGgctagctacaacgaAACAGGTGG	96
AAACTTGGAggctagctacaacgaCACAAACAGG	97
CTGATTAAAggctagctacaacgaTTGGACCAC	98
TGGTGCTGAggctagctacaacgaTAACTTGG	99
TGATGATCAGgctagctacaacgaCTCTGTCTG	100
TGGTGATGAggctagctacaacgaCATCTCTGT	101
GCTTGGTGAggctagctacaacgaGATCATCTC	102
ATGGGAACAggctagctacaacgaCCGCCGTCC	103
GAATGGGAAGgctagctacaacgaATCCGCCGT	104
TGACAGGAAGgctagctacaacgaGGGAACATC	105
AGTAAATGAggctagctacaacgaAGGAATGGG	106
CACAGTAAAggctagctacaacgaGACAGGAAT	107
GCCCCGCCAggctagctacaacgaAGTAAATGA	108
CCACAAACAggctagctacaacgaCCTGTAGTG	109
GTCCACAAAggctagctacaacgaATCCTGTAG	110
CCACGTCCAggctagctacaacgaAAACATCCT	111
CCAAGACCAggctagctacaacgaGTCACAAA	112
CCACCAAGAggctagctacaacgaCACGTCCAC	113
GCTGGTCCAggctagctacaacgaCAAGACCAC	114

10

TABLE 1-continued

Dnzyme sequence	Sequence identifier
GCTCTGGTAggctagctacaacgaCGCCAGTGG	115
CTGCACCCAggctagctacaacgaTTGCCGCTC	116
CACACTGCAGgctagctacaacgaCCACTTGCC	117
CTTCCACAggctagctacaacgaTGACCCAC	118
GCCTTTCCAggctagctacaacgaACTGCACCC	119
TTCTGGCAGgctagctacaacgaGCTGCCCTC	120
GTGGACGTAggctagctacaacgaAGGCGGTTT	121
CCGGGTGGAggctagctacaacgaGTACAGGCG	122
CCTGGCGCAGgctagctacaacgaCCAGTGCGC	123
CAATGAAAggctagctacaacgaTTCCTGGCG	124
TTTCCCAAAGgctagctacaacgaGAAACTTCC	125
ATTGTTGGAggctagctacaacgaGCCCCCTTG	126
TGGGTCACAggctagctacaacgaTGTTGACG	127
TCTGGGTCAggctagctacaacgaATTGTTGGA	128
GCACAATCAGgctagctacaacgaCTGGGTAC	129
GGAGCACAAGgctagctacaacgaCATCTGGGT	130
ACTGGAGCAGgctagctacaacgaAATCATCTG	131
ATGGAGGGAggctagctacaacgaTGGAGCACA	132
TGGTACTTAggctagctacaacgaGGAGGGACT	133
GGGCTGGTAggctagctacaacgaTTATGGAGG	134
TCAACGATAggctagctacaacgaGCAGCCGGG	135
CCTCAACGAggctagctacaacgaATGCAGCCG	136
TCACCTCAAaggctagctacaacgaGATATGCAG	137
CGTCGTTCAggctagctacaacgaCTCAACGAT	138
GTAAAGATAggctagctacaacgaGCGTGTGG	139
AAGTAAAGAggctagctacaacgaATGCGTGTT	140
GGCAATGAAggctagctacaacgaTGGGTTTCT	141
TCACGGCAAggctagctacaacgaGAAGTGGGT	142
AGGCAGTCAggctagctacaacgaGGCAATGAA	143
ATCTCGGCAGgctagctacaacgaTCTGGTAGG	144
GCTGAGTAAggctagctacaacgaCTCGGCATT	145
TATTATCAAaggctagctacaacgaTTTCAGCTG	146
GGGTATTAggctagctacaacgaCAATTTTCA	147
AAGGGGTTAggctagctacaacgaTATCAATTT	148
CTCCCGGAaggctagctacaacgaCCTTTGGCA	149
GTACATGGAggctagctacaacgaTCAAAGTTC	150
AACTGGGTAggctagctacaacgaGGCAGAATA	151

65 Dnazymes as used herein can be conjugated to a carrier molecule for stability or targeting purposes. Nanoparticles,

micelles, liposomes, polymer complexes, may be used to increase in vivo stability of the dnzyme(s). For example, gold nanoparticles may be conjugated to dnzymes as described herein. In some embodiments, the nanoparticles are further complexed to a targeting moiety, e.g., such as an antibody, monoclonal antibody, targeting molecule, or ligand for a receptor on the target cell(s) or tissue. In some embodiments, the dnzyme is conjugated to a targeting moiety via a linker.

In some embodiments, non-naturally occurring dnzymes are synthesized and screened for GATA3 inhibitory activity in vitro using established methods as described, e.g., in U.S. Pat. No. 9,797,888.

Methods of Treatment

Some embodiments herein are directed to the use of a therapeutic agent comprising one or more GATA3 inhibitors for the treatment of insulin resistance, or related disorders or diseases, including, but not limited to obesity, hypertension, dyslipidemia and type 2 diabetes in an individual that has been diagnosed as having insulin resistance or Type II diabetes. Some embodiments are directed to the restoration of insulin sensitivity by administering one or more GATA3 inhibitors.

Some embodiments herein are directed to the use of a therapeutic agent or pharmaceutical formulation comprising one or more GATA3 inhibitors to reverse, delay or prevent onset of insulin resistance or Type II diabetes in an individual that is at risk for developing insulin resistance or Type II diabetes. Some embodiments are directed to the administration of a pharmaceutical formulation comprising one or more GATA3 inhibitors to reduce risk of developing type II diabetes in an individual. In some embodiments, GATA3 inhibitors are administered to reduce or eliminate the phenotypic and/or metabolic symptoms characteristic of insulin resistance or a disorder or disease associated with it.

In other embodiments, an agent comprising one or more GATA3 inhibitors is administered locally to a tissue, or a site adjacent to said tissue, in order to increase the adipogenic capacity in said tissue. In some embodiments, localized exposure to a composition comprising a GATA3 inhibitor results in a decrease in GATA3 levels in said tissue. In some embodiments the administration of the GATA3 inhibitor(s) will result in an increase in adipogenesis in said tissue, thereby locally increasing fat deposits. In some embodiments, application of one or more GATA3 inhibitors locally results in a conversion of stem cells or preadipocytes into adipocytes at the site of application or adjacent tissues. In some embodiments, localized exposure to a formulation comprising a GATA3 inhibitor results in an increase in the number or % of adipocytes in said tissue.

In some embodiments, exposure to an effective amount of a composition comprising one or more GATA3 inhibitor(s) results in a decrease in adipocyte volume in visceral fat deposits and an increase in adipocyte volume in subcutaneous fat deposits.

In some embodiments, the administration of the GATA3 inhibitor(s) for stimulation of conversion of stem cells or preadipocytes into adipocytes is to achieve a desired cosmetic effect. In some embodiments a cosmetically effective amount of the GATA3 is administered to an individual. In some embodiments, inhibition of GATA3 in a tissue results in increased levels of adipocytes in subcutaneous tissues with low levels of adipocytes. In some embodiments, the increased levels of adipocytes in a tissue are a result of GATA3 inhibition-based adipogenesis. In some embodiments, cosmetic formulations comprising one or more GATA3 inhibitors are administered at or adjacent to the

breast(s), thigh(s) or buttock(s) in order to stimulate adipogenesis in said thighs or buttocks. In some embodiments, cosmetic formulations comprising one or more GATA3 inhibitors are applied to the skin of the extremities or to the subcutaneous tissue of the extremities in order to increase adipogenesis in said subcutaneous tissues of the extremities. In some embodiments the extremities are the hands or feet. In other embodiments, formulations comprising one or more GATA3 inhibitors are administered to the subcutaneous tissues in the backs of the hands, or neck to stimulate adipogenesis in said subcutaneous tissues to alleviate fat loss in those areas due to aging or weight loss.

Patients

In some embodiments, pharmaceutical formulations comprising one or more GATA3 inhibitors are administered to individuals suffering a metabolic disorder selected from the group consisting of elevated blood glucose (e.g., reduced ability to normalize glucose), impaired glucose tolerance, insulin resistance, type II diabetes, obesity, elevated percent body fat, and fatty liver (hepatic steatosis). Said individuals may be treated by the compositions and methods described herein. In some embodiments, the individual is a mammal. In some embodiments, the mammal is a human.

In other embodiments, individuals can be administered a composition comprising a GATA3 inhibitor for cosmetic purposes, i.e., a cosmetically effective amount of a composition comprising one or more GATA3 inhibitors to stimulate adipogenesis or to stimulate fat accumulation in a localized area. In some embodiments, the individual is a mammal. In some embodiments, the mammal is a human.

In some embodiments, patients are administered a GATA3 inhibitor in order to stimulate fat redistribution from a visceral deposit in the individual to a subcutaneous deposit in the thigh(s), buttock(s), extremities, neck, chest or face. In some embodiments the individual is a mammal. In some embodiments the mammal is a human.

In some embodiments, the visceral fat deposit is adjacent to the liver, stomach, or intestines. In some embodiments, the visceral fat deposit is the omentum.

Dosage Forms

In some embodiments described herein, the pharmaceutical compositions comprising a GATA3 inhibitor may be admixed with any suitable binder(s), lubricant(s), suspending agent(s), coating agent(s), solubilizing agent(s), and combinations thereof.

Solid oral dosage forms of pharmaceutical compositions, such as, tablets or capsules, comprising for example, dnzyme-based GATA3 inhibitors, can be administered in at least one dosage form at a time, as appropriate. These compositions can further contain excipients such as binders (e.g., hydroxypropylmethylcellulose, polyvinyl pyrrolidone, other cellulosic materials and starch), diluents (e.g., lactose and other sugars, starch, dicalcium phosphate and cellulosic materials), disintegrating agents (e.g., starch polymers and cellulosic materials) and lubricating agents (e.g., stearates and talc). In some embodiments, the GATA3 inhibitor based pharmaceutical composition is in a sustained release or extended release formulation. Additional oral forms in which the present inventive compounds may be administered include elixirs, solutions, syrups, and suspensions; each optionally containing flavoring agents and coloring agents.

Alternatively, compositions as described herein can be administered by inhalation (intratracheal or intranasal) or in the form of a suppository or pessary.

In some embodiments, the therapeutic, pharmaceutical, or cosmetic formulations comprising one or more GATA3

inhibitors are formulated for topical application and are in the form of a lotion, solution, cream, ointment or dusting powder.

Creams comprising one or more GATA3 inhibitors as active ingredients can comprise a three phase system of an aqueous, an oily, and an emulsifying phase. In some embodiments, the emulsion of oil and water is in approximately equal proportions. Additionally, certain embodiments are directed to "augmented" creams, meaning they are more potent and penetrate the skin deeper. In some embodiments, creams comprising one or more GATA3 inhibitors can further comprise one or more cosmetically and/or pharmaceutically acceptable emulsifying agents, penetrating agent, permeation enhancers, active agent carriers and preservatives.

For topical administration, some embodiments can comprise, for example, gels and solutions, and can contain excipients such as solubilizers, permeation enhancers (e.g., fatty acids, fatty acid esters, fatty alcohols and amino acids), and hydrophilic polymers (e.g., polycarbophil and polyvinylpyrrolidone). In the preferred embodiment, the pharmaceutically acceptable carrier is a liposome or a biodegradable polymer.

Transdermal delivery systems further include patches, gels, tapes and creams, and can contain excipients such as solubilizers, permeation enhancers (e.g., fatty acids, fatty acid esters, fatty alcohols and amino acids), hydrophilic polymers (e.g., polycarbophil and polyvinylpyrrolidone), and adhesives and tackifiers (e.g., polyisobutylenes, silicone-based adhesives, acrylates and polybutene). GATA3 inhibitors can be incorporated at a concentration of between about 0.01% and about 10% by weight, between about 0.05% and 10% by weight, between about 0.1 and about 10% by weight, between about 0.5% and about 10% by weight, between about 1 and about 10% by weight of the pharmaceutical formulation. In a preferred embodiment, said GATA3 inhibitors are encased in liposomes for delivery through the skin into the subcutaneous tissue. For example, the pharmaceutical composition can further cream, into an ointment comprising, consisting of, and/or consisting essentially of a wax or soft paraffin base together with any stabilizers and preservatives as may be required. An alternative means of administration includes transdermal administration by using a skin or transdermal patch.

The pharmaceutical compositions of the present invention (as well as the compounds of the present invention alone) can also be injected parenterally, for example, subcutaneously, intracavernosally, intravenously, intramuscularly, subcutaneously, intradermally, or intrathecally. In some embodiments, the compositions will also include at least one of a suitable carrier, a suitable excipient, and a suitable diluent.

In some embodiments described herein, the GATA3 inhibitors are bound or linked to a carrier for more efficient delivery of the inhibitor to the target tissue(s) or cells(s). GATA3 inhibitors as described herein can be bound to nanoparticles, micelles, or liposomes to improve specificity of delivery, or increase durability of said inhibitor in the body.

For parenteral administration, the pharmaceutical compositions of some embodiments are best used in the form of a sterile aqueous solution that may contain other substances, for example, enough salts and monosaccharides to make the solution isotonic with the tissue into which it is administered.

For buccal or sublingual administration, some embodiments as described herein may be administered in the form of tablets or lozenges, which can be formulated in a conventional manner.

In embodiments comprising topical administration of a GATA3 inhibitor comprising one or more molecules greater than 1000 daltons in size, such as one or more siRNAs, dnazymes, or antibodies, said administration can be done using microneedling, microporation, electroporation, iontophoresis, sonophoresis, or via passive administration using polymeric nanoparticles, liposomes, peptides, dendrimers or other carrier agents.

Therapeutic Dosage

Some embodiments are directed to methods of treating insulin resistance or a related disease or disorder in an individual by administration of a composition comprising one or more dnazymes with GATA3 inhibitory activity, wherein the dnzyme(s) are at a concentration between 1 ng/kg-10 mg/kg body weight, between 1 µg-10 mg/kg body weight, between 100 µg-10 mg/kg body weight, between 1 mg-10 mg/kg body weight, between 2.5 ng/kg-10 mg/kg body weight, between 2.5 µg-10 mg/kg body weight, between 250 µg-10 mg/kg body weight, between 2.5 mg-10 mg/kg body weight, between 5 ng/kg-10 mg/kg body weight, between 5 µg-10 mg/kg body weight, between 50 µg-10 mg/kg body weight, or between 500 or between 1 mg-10 mg/kg body weight.

Some embodiments are directed to methods of treating insulin resistance or a related disease or disorder in an individual by administration of a composition comprising one or more dnazymes with GATA3 inhibitory activity, wherein the dnzyme(s) are at a concentration in the range of between 0.01 and 10%, between 0.1 and 10%, between 1 and 10%, between 0.05 and 10%, between 0.1 and 10%, between 0.05 and 10%, between 0.0025 and 10%, between 0.025 and 10%, between 0.25 and 10%, between 2.50 and 10%, between 0.01 and 0.05%, between 0.01 and 0.5%, or between 0.01 and 5% of said composition.

Some embodiments described herein are directed to treatment regimens wherein said therapeutic composition is administered 1x, 2x, 3x, 4x, or 5x daily.

Cosmetic Dosage

Some embodiments are directed to cosmetic methods for inducing adipogenesis in an individual by administration of a composition comprising one or more dnazymes with GATA3 inhibitory activity. In some embodiments the cosmetic composition is a cream, gel or ointment. In some embodiments the composition comprises dnzyme(s) with GATA3-inhibitory activity at a concentration in the range of between 0.01 and 10%, between 0.1 and 10%, between 1 and 10%, 0.01 and 10%, between 0.1 and 10%, between 1 and 10%, between 0.0025 and 10%, between 0.025 and 10%, between 0.25 and 10%, between 2.50 and 10%, between 0.01 and 0.05%, between 0.01 and 0.5%, or between 0.01 and 5% of said cosmetic composition.

Those embodiments directed to methods for effecting adipogenesis in target tissues, e.g., in the subcutaneous fat deposits of breast(s), thigh(s), and buttock(s), may require one or more physically invasive procedures comprising microneedling, microporation, electroporation, iontophoresis, or sonophoresis to achieve the desired results, i.e., increasing the size of fat deposits in said tissues.

In those embodiments using passive methods to deliver the GATA3 inhibitor(s) of the invention to target tissues, application of cosmetic compositions comprising the GATA3 inhibitor(s) and carrier agents may require repeated application. For example, passive administration of the

GATA3 inhibitors for cosmetic purposes may require daily, semi-daily, weekly, or semiweekly for a period of between 1 day and 1 year.

Combination Therapy

Some embodiments are directed to the use of one or more GATA3 inhibitors in combination therapy for the treatment of insulin resistance. For example, GATA3 inhibitors of the invention may be administered simultaneously or sequentially with metabolic disease-based drugs including, but not limited to Metformin, Glucophage, Glucophage XR, Formetam, Glumetza, Riomet), insulin, Humulin R, or Novolin R for the restoration of insulin sensitivity.

EXAMPLES

Example 1: Expression of Adipogenic Factors in Insulin Sensitive and Insulin Resistant Individuals

Materials and Methods

Cohort: Recruitment criteria of participants were previously described (Almuraikhy et al., *Diabetologia*, 59(11): 2406-2416 (2016)). Protocols were approved by the Institutional Review Board of ADLQ (X2017000224). Data from paired subcutaneous (SC) and omental (OM) adipose tissues from 10 consented obese females undergoing bariatric surgery at Al-Emadi hospital (Doha, Qatar) were used for this study. Insulin resistance was assessed by the homeostatic model assessment (HOMA-IR) (Grundy et al., *Arterioscler Thromb Vasc Biol.*, 24(2):e13-8 (2004)), using 30th percentile (HOMAIR=2.4) as a threshold point. Accordingly, subjects were dichotomized into IS (HOMA-IR<2.4, n=5 and IR (HOMA-IR>2.4, n=5).

Secreted cytokine profiling: Profiling of secreted cytokines from SC preadipocytes from 5-6 participants was performed using media supernatants collected following completion of differentiation. Accumulated levels of secreted IL-6 and IL-10 in the last 4 days before staining were measured using Inflammatory Cytokine Human Magnetic 5-Plex (Life Technologies) according to manufacturer's instructions and assessed by Luminex Flexmap 3D using xPONENT 4.2 software.

3T3L-1 cell culture and differentiation: Mouse 3T3L-1 preadipocytes were cultured in DMEM growth media (Gibco) supplemented with 10% heat-inactivated fetal bovine serum (Sigma) and 1% antibiotics (Gibco). After reaching confluency, cells were induced to differentiate (day 0) using a growth medium supplemented with 10 µg/ml insulin, 0.5 mM 1-methyl-3-isobutyl-xanthine (IBMX) and 1 µM dexamethasone. On day 3, the medium was changed to post-differentiation medium supplemented only with 10 µg/ml insulin until day 10. To induce insulin resistance, cells were differentiated in the presence of 10 µM of 4-hydroxynonenal (4HNE) as previously described (Elkayress et al., *Free Radic. Biol. Med.* 104:129-137 (2017)).

Targeting GATA3 expression: GATA3 mRNA specific DNAzyme sequence (hgd40: GTGGATGGAGGCTAGC-TACAACGAGTCTTGGAG)(SEQ ID NO:1) and the control scrambled DNAzyme sequence (ODNg: 5'-TGT-GATGTAGGCTAGCTACAACGACAGATGGAA)(SEQ ID NO:152) were obtained from Integrated DNA Technologies (IDT). 3T3L-1 cells, seeded in 24 well plate (42000 cells/well) one day before transfection, were transfected with DNAzyme complexed with Lipofectamine 3000 transfection reagent (Thermo Fisher Scientific) in Opti-MEM media (Gibco) with a final concentration of DNAzyme of 1 ng/µL according to manufacturer's instructions. Transfec-

tion was carried out for 6-8 hours followed by replacing media with adipogenic differentiation media.

Assessment of adipogenesis: At day 11, differentiated adipocytes were stained with oil red O (ORO) for assessment of adipogenesis as described previously (Krause et al., *Adipocyte*, (2016)). Briefly, differentiated cells were fixed with 4% paraformaldehyde for 15 minutes at room temperature, washed with 1×PBS and incubated with ORO working solution for 30 min at room temperature. The wells were then washed 5 times with water, followed by dye elution with 100% 2-propanol. The absorption of resulting color was measured using EPOCH2 microplate reader from BioTek at 510 nm according to manufacturer's instructions.

Western Blotting: Cells were washed once in PBS and lysed directly in RIPA lysis buffer (150 mM NaCl, 50 mM Tris pH 7.5, 0.1% SDS, 1 mM EDTA, 1 mM EGTA, 0.5% NaDoc (Sigma), 1% Triton X-100) supplemented with protease inhibitor cocktail (Sigma), 10 mM sodium fluoride (Sigma), 1 mM sodium orthovanadate (Sigma), 1 mM PMSF (Sigma), 5 mM benzamidine (Sigma), 20 µg/ml calpain inhibitor (Sigma), 5 mM nicotinamide (Sigma), and 3 mM trichostatin A (InvivoGen). Protein lysates were homogenized by a sonicator, centrifuged at 15,000 g for 10 min at 4° C. and supernatants were collected for protein estimation using DC Protein Assay (Bio-Rad) with CLARIOstar microplate reader (BMG LABTECH). 50-100 µg of whole-cell extracts per sample were separated on SDS-PAGE and transferred to PVDF membrane (Bio-Rad). The blots were probed with the following antibodies: GATA-3 (D13C9; Cell Signaling Technology, RRID: AB 2798212), GAPDH (14C10; Cell Signaling Technology, RRID: AB 561053). Signals were detected in the presence of Super-Signal™ West Dura Extended Duration Substrate (Thermo Fisher Scientific) using ChemiDoc™ MP imaging system (Biorad). Band intensities were quantified using ImageJ software.

Assessment of gene Expression: RNA from differentiated adipocytes was extracted using TRIzol method (Invitrogen) according to manufacturer's instructions. Three µg of the resulting RNA was used for First-strand cDNA synthesis using Superscript III first strand synthesis super mix kit (Invitrogen) according to manufacturer's instructions. Real-time PCR was carried for gene expression analysis using the produced cDNA with the primers listed in Table 2 using 7500 Real Time PCR System from Applied Biosystem. The PCR conditions were as follows: 1 cycle of 95° C. for 10 min, 45 cycles of 95° C. for 15 s, 55° C. for 40 s and 72° C. for 30 cycles and finally 60° C. for 15 s. Real-time PCR was carried out in triplicate and the GAPDH was used as a housekeeping gene for normalization of the amplified signals of the target genes. The data analysis was performed using the $\Delta\Delta C_t$ based calculation. For measuring the effect of GATA3 inhibition on insulin resistance-related genes, RT² Profiler™ PCR Array Mouse Insulin Resistance (PAMNI-156Z) was used according to the manufacturer's protocol (QIAGEN). The array contains C-C chemokine receptor type 5 (Ccr5), Janus Kinase 2 (Jak2), caspase 1 (Casp1), Sterol regulatory element-binding transcription factor 1 (Srebf1), Glucuronidase beta (Gusb), C-C chemokine receptor type 4 (Ccr4), leptin receptor (Lepr) and Peroxisome proliferator-activated receptor alpha (Ppara). The PCR experiments were performed using QuantStudio 6 Flex from Applied Biosystem and data analyzed using the $\Delta\Delta C_t$ -based calculations.

TABLE 2

A list of primers sequence.		
Gene	Primers sequences (5' to 3')	SEQ ID NO:
GATA3	f: GAACCGGCCCTTATCAAG	153
	r: ACAGTTCGCGCAGGATGTC	154

(58±26 ng/ml) and hyperinsulinemia (10.6±5.4 mIU/L) and were dichotomized into IS and IR groups based on their HOMA-IR index.

Insulin resistance was assessed by the homeostatic model assessment (HOMA-IR) ((Grundy et al., *Arterioscler Thromb Vasc Biol.*, 24(2):e13-8 (2004)) using 30th percentile (HOMA-IR=2.4) as a threshold point. Subjects were dichotomized into IS (HOMA-IR<2.4, n=5) and IR (HOMA-IR>2.4, n=5). Table 3 provides characteristics of the participant groups.

TABLE 3

Variables	Cohort (n = 10)	IS (N = 5)	IR (N = 5)	p value
Age (year)	33.1 (11.6)	33.6 (13.1)	29.8 (10.2)	0.62
BMI (kg · m ⁻²)	44 (5.2)	42.5 (2)	44.8 (7.6)	0.52
SBP (mmHg)	118.5 (13.5)	121.2 (9.6)	113.6 (17.2)	0.41
DBP (mmHg)	67.7 (12.2)	66.6 (14.0)	68.00 (13)	0.87
MAP	84.7 (12.1)	84.8 (12.2)	83.2 (12.3)	0.86
Cholesterol (mmol/L)	4.8 (0.84)	4.3 (0.92)	5.1 (0.65)	0.16
LDL (mmol/L)	2.9 (0.81)	2.7 (1.03)	3 (0.65)	0.61
Triglyceride (mmol/L)	1.1 (0.35)	1.01 (0.3)	1.1 (0.44)	0.69
Triglyceride/HDL	2.02 (1.3)	2.4 (1.9)	1.6 (0.73)	0.4
HDLC (mmol/L)	1.4 (0.39)	1.2 (0.35)	1.6 (0.36)	0.08
Leptin (ng/ml)	58 (26)	60 (35)	57.5 (22.2)	0.9
IL-6 (pg/ml)	3.6 (2)	3 (1.1)	4.1 (2.5)	0.34
FBG (mmol/L)	7.3 (4.4)	5.2 (0.54)	6.8 (2.6)	0.24
Insulin (mIU/L)	10.6 (5.4)	6.8 (4.3)	13.8 (4.7)	0.03
HOMA-IR	3.4 (2.4)	1.6 (0.9)	4 (1.4)	0.009
Albumin (g/L)	42 (4.8)	43.6 (4)	41.7 (5)	0.5
ALP (IU/L)	72 (22.3)	83.2 (25.3)	59.3 (14)	0.078
ALT (IU/L)	27.1 (18.4)	22.8 (8.6)	31.2 (25.4)	0.5
AST (IU/L)	20.8 (10.9)	18.8 (3.4)	23.7 (14.6)	0.5
Bilirubin (μmol/L)	8.5 (3.6)	10.8 (3)	7.2 (3.4)	0.1

Mediators of metabolic syndrome were measured in insulin sensitive (IS) and insulin resistant (IR) including body

TABLE 2-continued

A list of primers sequence.		
Gene	Primers sequences (5' to 3')	SEQ ID NO:
GAPDH	f: AGGTCGGTGTGAACGGATTG	155
	r: TGTAGACCATGTAGTTGAGGTCA	156
IL-6	f: TAGTCCTTCTACCCCAATTTCC	157
	r: TTGGTCCTTAGCCACTCCTTC	158
IL-10	f: GCTCTTACTGACTGGCATGAG	159
	r: CGCAGCTCTAGGAGCATGTG	160

f: forward, r: reverse, GATA3: GATA binding protein 3, GAPDH: Glyceraldehyde 3-phosphate dehydrogenase, IL6: interleukin 6 and IL10: interleukin 10.

Insulin signaling analysis: The effect of GATA3 inhibition on insulin resistance biomarkers was quantified in equal volumes of lysates prepared from GATA3 DNzyme transfected 3T3L-1 cells using Bio-Plex Pro Mouse Diabetes 8-Plex Assay (171F7001M, BIO-RAD), following manufacturer's instructions and assessed by Luminex 200 using xPONENT 4.2 software.

Statistical analysis: T-test and Wilcoxon-Mann-Whitney test were performed as appropriate using IBM SPSS statistics 21 (Armonk, NY, USA). Non-parametric tests were used for comparing ordinal or non-normal variables. Significance was defined as p≤0.05.

Individuals were premenopausal (33.1±years of age) with a weight range of (43.9±5.2 Kg/m²). Participants showed hypercholesterolemia (4.8±0.84 mM/L), hyperleptinemia

Profiling of secreted cytokines from SC preadipocytes from 5-6 participants was performed using media supernatants collected following completion of differentiation. Accumulated levels of secreted IL-6 and IL113 in the last 4 days before staining was assessed using Inflammatory Cytokine Human Magnetic 5-Plex (Life Technologies) according to manufacturer's instructions and assessed using Luminex Flexmap 3D using xPONENT 4.2 software.

Adipose tissue samples taken from insulin sensitive and insulin resistant individuals and mRNA levels were also assessed using RT-PCR using RT2 Profiler™ PCR Array Human Adipogenesis, Catalog No. PAHS-049Z(Qiagen).

GATA-3 expression was quantified in SC and OM adipose tissues and in preadipocyte cultures expanded from SC adipose tissues from IS and IR individuals. Western blotting data indicated higher expression of GATA3 in OM compared to SC adipose tissues, regardless of the insulin resistance status of tissue donors (FIG. 1). RT-PCR results showed higher expression of GATA3 in preadipocytes and differentiating preadipocytes from IR obese individuals compared to IS counterparts (FIG. 2). Interestingly, there was a significant positive correlation between GATA3 expression and secreted IL-6 and IL-beta from SC and OM differentiated adipocytes as presented as mean±SEM. *P<0.05, paired and independent sample t-test. (FIG. 3).

Example 2: Effect of GATA3 Inhibition on 3T3L-1 Preadipocyte Adipogenic Differentiation

To determine whether inhibition of GATA3 could affect the adipogenic capacity of murine 3T3L-1 cells in vitro, GATA3 expression was assessed following transfection with hgd40 DNzyme targeting GATA3 mRNA in the presence or absence of the anti-adipogenic reactive aldehyde 4HNE.

Mouse 3T3L1 preadipocytes were cultured in DMEM growth media (Gibco) supplemented with 10% heat-inactivated fetal bovine serum (Sigma) and 1% antibiotics (Gibco). After reaching confluency, cells were induced to differentiate (day 0) using a growth medium supplemented with 10 µg/ml insulin, 0.5 mM 1-methyl-3-isobutyl-xanthine (IBBMX) and 1 µM dexamethasone. On day 3, the medium was changed to post-differentiation medium supplemented only with 10 µg/ml insulin until day 10. To induce insulin resistance, cells were differentiated in the presence of 10 µM of 4-hydroxynonenal (4HNE).

DNA sequences (hgd40: 5'-GTGGATGGAGGCTAGC-TACAACGAGTCTTGGAG)(SEQ ID NO:1) and the control scrambled DNAzyme sequence (ODNg: 5'-TGT-GATGTAGGCTAGCT-ACAACGACAGATGGAA)(SEQ ID NO:152) were obtained from Integrated DNA Technologies (IDT). 3T3L1 cells, seeded in 24 well plate (42000 cells/well) one day before transfection, were transfected with DNAzyme complexed with Lipofectamine 3000 transfection reagent (Thermo Fisher Scientific) in Opti-MEM media (Gibco) with a final concentration of DNAzyme of 1 ng/µL according to manufacturer's instructions. Transfection was carried out for 6-8 hours followed by replacing media with adipogenic differentiation media.

GATA3 expression was down-regulated in differentiated adipocytes (FIG. 4), and triggered higher adipogenic capacity of 3T3L1 preadipocytes (FIG. 5). Additionally, co-treatment of 3T3L1 with the anti-adipogenic reactive aldehyde 4HNE and GATA3 inhibitor (hgd40) rescued adipogenesis compared with cells treated with 4HNE alone (FIG. 8). Additionally, co-treatment of cells with 4HNE and hgd40 improved their insulin sensitivity by increasing the expression of genes associated with insulin sensitivity, including Ccr4, Gusb, Lepr and Ppara, and decreasing the expression of genes associated with insulin resistance such as Jak2, Casp1, Srebf1 and Ccr5 (FIG. 6).

The superior adipogenic capacity in metabolically healthy obese (MHO) individuals MHO group is found to be medi-

ated, at least in part, by lower levels of the inflammatory mediator interleukin-6 (IL-6) (Almuraikhy et al., *Diabetologia*, 59(11):2406-2416 (2016)), reactive aldehyde 4-hydroxynonenal (4-HNE) (Elrayess et al., *Free Radic Biol Med*, 104:129-137 (2017)), the environmental pollutants polybrominated diphenyl ethers (Helaleh et al., *Chemosphere*, 209:268-276 (2018)), and the combined treatment of metformin and insulin (Jaganjac et al., *Redox Biol*, 12:483-490 (2017) and Al-Sulaiti et al., Intech Open, DOI: 10.5772, (2019)). As seen in FIG. 3, lower panel, preadipocytes treated with the dnzyme-based GATA3 inhibitor had reversal of the reactive aldehyde 4-hydroxynonenal (4-HNE) induced impairment of adipogenesis in murine 3T3L1 cells as well as increased adipogenic capacity. As seen in FIG. 4, upper panel, treatment of primary 3T3L1 mouse preadipocytes with the dnzyme-based GATA3 inhibitor showed similar results. FIGS. 3 and 6 show the correlation between GATA3 mRNA of secreted inflammatory cytokines IL1β, IL-6, and the correlation between GATA3 and circulating IL-6 protein levels, respectively.

Example 3: Effect of GAT-3 on Insulin Signaling Genes and Insulin Sensitivity

To investigate whether GATA3 expression affects inflammatory markers associated with obesity and insulin resistance, the impact of GATA3 inhibition on the expression of pro-inflammatory IL6 and anti-inflammatory IL10 was investigated. Results indicated a significant reduction in IL6 expression and an increase in IL10 expression following GATA3 inhibition (FIG. 7). Similarly, the effect of GATA3 inhibition on expression of insulin resistance-related proteins was assessed in 3T3L1. Data indicated a significant reduction in PAI-1, resistin and glucagon in response to GATA3 inhibition (FIG. 8).

All referenced disclosed herein are incorporated by reference in their entirety.

SEQUENCE LISTING

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misc_feature          1..33
                        note = Dnzyme
source                1..33
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 1
gtggatggag gctagctaca acgagtcttg gag                               33

SEQ ID NO: 2          moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                        note = Dnzyme
source                1..33
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 2
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SEQ ID NO: 3          moltype = DNA length = 33
FEATURE              Location/Qualifiers
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                        note = Dnzyme
source                1..33
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 3
tcggtcagag gctagctaca acgatgcgtt gct                               33
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-continued

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SEQ ID NO: 4      moltype = DNA  length = 33
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source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 4
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SEQ ID NO: 5      moltype = DNA  length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 5
ggcgtagcgag gctagctaca acgagacctg ctc      33

SEQ ID NO: 6      moltype = DNA  length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 6
ctcggtcgag gctagctaca acgactgggt agc      33

SEQ ID NO: 7      moltype = DNA  length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 7
tcctctgcag gctagctaca acgacgggt cct      33

SEQ ID NO: 8      moltype = DNA  length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 8
actctgcaag gctagctaca acgatctgcg agc      33

SEQ ID NO: 9      moltype = DNA  length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 9
ggcgtagcgag gctagctaca acgatctgca att      33

SEQ ID NO: 10     moltype = DNA  length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 10
aagggcgag gctagctaca acgagactct gca      33

SEQ ID NO: 11     moltype = DNA  length = 33
FEATURE          Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 11

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aaaacgggag gctagctaca acgacaggtt gta 33

SEQ ID NO: 12      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 12
agaataaaaag gctagctaca acgaggggacc agg 33

SEQ ID NO: 13      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 13
atggcagaag gctagctaca acgaaaaacg gga 33

SEQ ID NO: 14      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 14
atccaaaaag gctagctaca acgatgggta tgg 33

SEQ ID NO: 15      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 15
aggggaagag gctagctaca acgaaaaaat cca 33

SEQ ID NO: 16      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 16
ttttaaaaag gctagctaca acgatatctt gga 33

SEQ ID NO: 17      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 17
gtggggggag gctagctaca acgaggggaag gct 33

SEQ ID NO: 18      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 18
gttgaatgag gctagctaca acgattgctt tcg 33

SEQ ID NO: 19      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 19
gtcgttgaag gctagctaca acgagatttg ctt                                     33

SEQ ID NO: 20      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 20
ggcccgaag gctagctaca acgaccgcgc gcg                                     33

SEQ ID NO: 21      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 21
tcacctccag gctagctaca acgaggcttc ggc                                     33

SEQ ID NO: 22      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 22
cgcgcgtcag gctagctaca acgactccat ggc                                     33

SEQ ID NO: 23      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 23
ggtggctcag gctagctaca acgaccagcg cgg                                     33

SEQ ID NO: 24      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 24
cgttgagcag gctagctaca acgaggcggg gtg                                     33

SEQ ID NO: 25      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 25
ccgcgtccag gctagctaca acgagtagga gtg                                     33

SEQ ID NO: 26      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 26
cagcgggtag gctagctaca acgatgcgc gcg                                     33

SEQ ID NO: 27      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA

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                                organism = synthetic construct
SEQUENCE: 27
gcacatccag gctagctaca acgactctc cgg                                     33

SEQ ID NO: 28                moltype = DNA  length = 33
FEATURE                      Location/Qualifiers
misc_feature                 1..33
                                note = Dnzyme
source                      1..33
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 28
aaaagcacag gctagctaca acgaccacct cct                                     33

SEQ ID NO: 29                moltype = DNA  length = 33
FEATURE                      Location/Qualifiers
misc_feature                 1..33
                                note = Dnzyme
source                      1..33
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 29
taaaaagcag gctagctaca acgaatccac ctc                                     33

SEQ ID NO: 30                moltype = DNA  length = 33
FEATURE                      Location/Qualifiers
misc_feature                 1..33
                                note = Dnzyme
source                      1..33
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 30
gaccgtcgag gctagctaca acgagttaaa aag                                     33

SEQ ID NO: 31                moltype = DNA  length = 33
FEATURE                      Location/Qualifiers
misc_feature                 1..33
                                note = Dnzyme
source                      1..33
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 31
ttgccttgag gctagctaca acgacgtcga tgt                                     33

SEQ ID NO: 32                moltype = DNA  length = 33
FEATURE                      Location/Qualifiers
misc_feature                 1..33
                                note = Dnzyme
source                      1..33
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 32
agggcgaggag gctagctaca acgagtgggt gcc                                     33

SEQ ID NO: 33                moltype = DNA  length = 33
FEATURE                      Location/Qualifiers
misc_feature                 1..33
                                note = Dnzyme
source                      1..33
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 33
tggccctgag gctagctaca acgacgagtt tcc                                     33

SEQ ID NO: 34                moltype = DNA  length = 33
FEATURE                      Location/Qualifiers
misc_feature                 1..33
                                note = Dnzyme
source                      1..33
                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 34
acctctgcag gctagctaca acgacgtggc cct                                     33

SEQ ID NO: 35                moltype = DNA  length = 33
FEATURE                      Location/Qualifiers
misc_feature                 1..33
                                note = Dnzyme
source                      1..33

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	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 35		
cggagggtag gctagctaca acgactctgc acc		33
SEQ ID NO: 36	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 36		
ggcggcacag gctagctaca acgactggct ccc		33
SEQ ID NO: 37	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 37		
cggggggcag gctagctaca acgaacctgg ctc		33
SEQ ID NO: 38	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 38		
agggatccag gctagctaca acgagaagca gag		33
SEQ ID NO: 39	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 39		
gggtagggag gctagctaca acgaccatga agc		33
SEQ ID NO: 40	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 40		
gggctgagag gctagctaca acgatccagg ggg		33
SEQ ID NO: 41	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 41		
gtggatggag gctagctaca acgagtcttg gag		33
SEQ ID NO: 42	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 42		
cgtggtggag gctagctaca acgaggacgt ctt		33
SEQ ID NO: 43	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	

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source          1..33
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 43
gggggtagag gctagctaca acgaggagag ggg                      33

SEQ ID NO: 44      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                 note = Dnzyme
source            1..33
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 44
ggaggaggag gctagctaca acgagaggcc ggg                      33

SEQ ID NO: 45      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                 note = Dnzyme
source            1..33
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 45
gcccccgag gctagctaca acgaaaggag gag                      33

SEQ ID NO: 46      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                 note = Dnzyme
source            1..33
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 46
cgggggagag gctagctaca acgagtcctt cgg                      33

SEQ ID NO: 47      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                 note = Dnzyme
source            1..33
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 47
ggacagcgag gctagctaca acgagggtcc ggg                      33

SEQ ID NO: 48      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                 note = Dnzyme
source            1..33
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 48
tggggtggag gctagctaca acgaagcgat ggg                      33

SEQ ID NO: 49      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                 note = Dnzyme
source            1..33
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 49
cttgaggcag gctagctaca acgatcttc tcg                      33

SEQ ID NO: 50      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                 note = Dnzyme
source            1..33
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 50
cacctggtag gctagctaca acgattgagg cac                      33

SEQ ID NO: 51      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33

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source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 51		
gcaggggcag gctagctaca acgactggta ctt		33
SEQ ID NO: 52	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 52		
ccagcttcag gctagctaca acgagctgtc ggg		33
SEQ ID NO: 53	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 53		
gtgggacgag gctagctaca acgatccagc ttc		33
SEQ ID NO: 54	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 54		
ggagtgggag gctagctaca acgagactcc agc		33
SEQ ID NO: 55	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 55		
atgctgccag gctagctaca acgagggagt ggg		33
SEQ ID NO: 56	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 56		
gggcgggtcag gctagctaca acgagctgcc acg		33
SEQ ID NO: 57	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 57		
gaggctccag gctagctaca acgaccaggg cgg		33
SEQ ID NO: 58	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 58		
gtgggtcgag gctagctaca acgagaggag gct		33
SEQ ID NO: 59	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	

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misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 59
aggtggtgag gctagctaca acgaggggtg gtg                               33

SEQ ID NO: 60      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 60
tggcttctag gctagctaca acgagccctc gtc                               33

SEQ ID NO: 61      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 61
gggctctgag gctagctaca acgagcctgg ctt                               33

SEQ ID NO: 62      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 62
gggaccccgag gctagctaca acgacgggagc ccg                               33

SEQ ID NO: 63      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 63
ggtgggggag gctagctaca acgacccacc gga                               33

SEQ ID NO: 64      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 64
ggcgggggag gctagctaca acgaccgagg gcc                               33

SEQ ID NO: 65      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 65
gggctgggag gctagctaca acgagggcag gga                               33

SEQ ID NO: 66      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 66
cgtcgaggag gctagctaca acgaccgcc ctc                               33

SEQ ID NO: 67      moltype = DNA length = 33

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FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 67		
gggctggcag gctagctaca acgacttccc gta		33
SEQ ID NO: 68	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 68		
cgatgcccg gctagctaca acgaccgggg cgg		33
SEQ ID NO: 69	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 69		
gctccacgag gctagctaca acgagcccat ccg		33
SEQ ID NO: 70	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 70		
cgggtccag gctagctaca acgagatgcc cat		33
SEQ ID NO: 71	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 71		
tctccgcaag gctagctaca acgaccggct cca		33
SEQ ID NO: 72	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 72		
ccgtcagcag gctagctaca acgagtctcc gca		33
SEQ ID NO: 73	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 73		
tccccggcag gctagctaca acgacggctc ggt		33
SEQ ID NO: 74	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 74		
actcgggcag gctagctaca acgagtaggg cgg		33

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SEQ ID NO: 75	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 75		
ggagctgtag gctagctaca acgacgggc acg		33
SEQ ID NO: 76	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 76		
ggacttgacg gctagctaca acgaccgaag ccg		33
SEQ ID NO: 77	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 77		
gggctggag gctagctaca acgattgcat ccg		33
SEQ ID NO: 78	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 78		
tgtgctggag gctagctaca acgacgggcc ttg		33
SEQ ID NO: 79	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 79		
gttcacacag gctagctaca acgatccctg cct		33
SEQ ID NO: 80	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 80		
cagttcacag gctagctaca acgaactccc tgc		33
SEQ ID NO: 81	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 81		
cacagttcag gctagctaca acgaacttc cct		33
SEQ ID NO: 82	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 82		
gttgccccag gctagctaca acgaagttca cac		33

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SEQ ID NO: 83      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 83
tcgccgccag gctagctaca acgaagtggg gtc          33

SEQ ID NO: 84      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 84
ccggtgccag gctagctaca acgactcgcc gcc          33

SEQ ID NO: 85      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 85
ggcgttgccag gctagctaca acgaaggtag tgt          33

SEQ ID NO: 86      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 86
tggcttctag gctagctaca acgagccctc gtc          33

SEQ ID NO: 87      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 87
cccccgccag gctagctaca acgagctcgt ccg          33

SEQ ID NO: 88      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 88
gtagggagag gctagctaca acgaccagg ctg          33

SEQ ID NO: 89      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 89
gggcgggcag gctagctaca acgacaaggc gcc          33

SEQ ID NO: 90      moltype = DNA  length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 90

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cgggaaggag gctagctaca acgatcgccc gcg	33
SEQ ID NO: 91	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = Dnzyme
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 91	
tagtcctcag gctagctaca acgagcggcc ccg	33
SEQ ID NO: 92	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = Dnzyme
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 92	
tccccgacag gctagctaca acgactccag tcc	33
SEQ ID NO: 93	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = Dnzyme
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 93	
tttccccgag gctagctaca acgaacctcc agt	33
SEQ ID NO: 94	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = Dnzyme
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 94	
tgagcgcgag gctagctaca acgacctcag ttt	33
SEQ ID NO: 95	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = Dnzyme
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 95	
ggaccacaag gctagctaca acgaaggtgg ttg	33
SEQ ID NO: 96	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = Dnzyme
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 96	
cttgaccag gctagctaca acgaaacagg tgg	33
SEQ ID NO: 97	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = Dnzyme
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 97	
aaacttgag gctagctaca acgacacaac agg	33
SEQ ID NO: 98	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = Dnzyme
source	1..33
	mol_type = other DNA
	organism = synthetic construct

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SEQUENCE: 98
ctgattaaag gctagctaca acgattggac cac                                     33

SEQ ID NO: 99      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source             1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 99
tggtgctgag gctagctaca acgataaact tgg                                     33

SEQ ID NO: 100     moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source             1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 100
tgatgacag gctagctaca acgactctgt ctg                                     33

SEQ ID NO: 101     moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source             1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 101
tggtgatgag gctagctaca acgacatctc tgt                                     33

SEQ ID NO: 102     moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source             1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 102
gcttggtgag gctagctaca acgagatcat ctc                                     33

SEQ ID NO: 103     moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source             1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 103
atgggaacag gctagctaca acgaccgccg tcc                                     33

SEQ ID NO: 104     moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source             1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 104
gaatgggaag gctagctaca acgaatccgc cgt                                     33

SEQ ID NO: 105     moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source             1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 105
tgacaggaag gctagctaca acgaggaac atc                                     33

SEQ ID NO: 106     moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature       1..33
                   note = Dnzyme
source             1..33
                   mol_type = other DNA

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                organism = synthetic construct
SEQUENCE: 106
agtaaatgag gctagctaca acgaaggaat ggg                                33

SEQ ID NO: 107      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Dnzyme
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 107
cacagtaaag gctagctaca acgagacagg aat                                33

SEQ ID NO: 108      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Dnzyme
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 108
gcccggccag gctagctaca acgaagtaaa tga                                33

SEQ ID NO: 109      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Dnzyme
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 109
ccacaaacag gctagctaca acgacctgta gtg                                33

SEQ ID NO: 110      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Dnzyme
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 110
gtccacaaaag gctagctaca acgaatcctg tag                                33

SEQ ID NO: 111      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Dnzyme
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 111
ccacgtccag gctagctaca acgaaaacat cct                                33

SEQ ID NO: 112      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Dnzyme
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 112
ccaagaccag gctagctaca acgagtccac aaa                                33

SEQ ID NO: 113      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Dnzyme
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 113
ccaccaagag gctagctaca acgacacgtc cac                                33

SEQ ID NO: 114      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = Dnzyme
source              1..33

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	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 114		
gctggtccag gctagctaca acgacaagac cac		33
SEQ ID NO: 115	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 115		
gctctggtag gctagctaca acgacgccag tgg		33
SEQ ID NO: 116	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 116		
ctgcaccag gctagctaca acgattgccg ctc		33
SEQ ID NO: 117	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 117		
cacactgcag gctagctaca acgaccactt gcc		33
SEQ ID NO: 118	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 118		
ctttccacag gctagctaca acgatgcacc cac		33
SEQ ID NO: 119	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 119		
gcctttccag gctagctaca acgaactgca ccc		33
SEQ ID NO: 120	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 120		
ttcctggcag gctagctaca acgagctgcc ctc		33
SEQ ID NO: 121	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 121		
gtggacgtag gctagctaca acgaaggcgg ttt		33
SEQ ID NO: 122	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	

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source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 122
cggggtggag gctagctaca acgagtacag gcg                                33

SEQ ID NO: 123         moltype = DNA length = 33
FEATURE               Location/Qualifiers
misc_feature          1..33
                      note = Dnzyme
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 123
cctggcgcag gctagctaca acgaccagtg cgc                                33

SEQ ID NO: 124         moltype = DNA length = 33
FEATURE               Location/Qualifiers
misc_feature          1..33
                      note = Dnzyme
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 124
caaatgaaag gctagctaca acgattcctg gcg                                33

SEQ ID NO: 125         moltype = DNA length = 33
FEATURE               Location/Qualifiers
misc_feature          1..33
                      note = Dnzyme
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 125
tttcccaaag gctagctaca acgagaaaact tcc                                33

SEQ ID NO: 126         moltype = DNA length = 33
FEATURE               Location/Qualifiers
misc_feature          1..33
                      note = Dnzyme
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 126
attgttggag gctagctaca acgagccccc ttg                                33

SEQ ID NO: 127         moltype = DNA length = 33
FEATURE               Location/Qualifiers
misc_feature          1..33
                      note = Dnzyme
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 127
tgggtcacag gctagctaca acgatgttg acg                                33

SEQ ID NO: 128         moltype = DNA length = 33
FEATURE               Location/Qualifiers
misc_feature          1..33
                      note = Dnzyme
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 128
tctgggtcag gctagctaca acgaattgtt gga                                33

SEQ ID NO: 129         moltype = DNA length = 33
FEATURE               Location/Qualifiers
misc_feature          1..33
                      note = Dnzyme
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 129
gcacaatcag gctagctaca acgactgggt cac                                33

SEQ ID NO: 130         moltype = DNA length = 33
FEATURE               Location/Qualifiers
misc_feature          1..33

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source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 130		
ggagcacaag gctagctaca acgacatctg ggt		33
SEQ ID NO: 131	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 131		
actggagcag gctagctaca acgaaatcat ctg		33
SEQ ID NO: 132	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 132		
atggaggagg gctagctaca acgatggagc aca		33
SEQ ID NO: 133	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 133		
tggactctag gctagctaca acgaggaggg act		33
SEQ ID NO: 134	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 134		
gggctggtag gctagctaca acgattatgg agg		33
SEQ ID NO: 135	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 135		
tcaacgtag gctagctaca acgagcagcc ggg		33
SEQ ID NO: 136	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 136		
cctcaacgag gctagctaca acgaatgcag ccg		33
SEQ ID NO: 137	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
source	note = Dnzyme 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 137		
tcacctcaag gctagctaca acgagatatg cag		33
SEQ ID NO: 138	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	

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misc_feature      1..33
                  note = Dnzyme
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 138
cgctggtcag gctagctaca acgactcaac gat                               33

SEQ ID NO: 139      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Dnzyme
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 139
gtaaagatag gctagctaca acgagcgtgt tgg                               33

SEQ ID NO: 140      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Dnzyme
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 140
aagtaaagag gctagctaca acgaatgcgt gtt                               33

SEQ ID NO: 141      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Dnzyme
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 141
ggcaatgaag gctagctaca acgatgggtt tct                               33

SEQ ID NO: 142      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Dnzyme
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 142
tcacggcaag gctagctaca acgagaactg ggt                               33

SEQ ID NO: 143      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Dnzyme
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 143
aggcagtcag gctagctaca acgaggcaat gaa                               33

SEQ ID NO: 144      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Dnzyme
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 144
atctcggcag gctagctaca acgatctggt agg                               33

SEQ ID NO: 145      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = Dnzyme
source              1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 145
gctgagtaag gctagctaca acgactcggc att                               33

SEQ ID NO: 146      moltype = DNA length = 33

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FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 146		
tattatcaag gctagctaca acgatttcag ctg		33
SEQ ID NO: 147	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 147		
gggttattag gctagctaca acgacaattt tca		33
SEQ ID NO: 148	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 148		
aaggggtag gctagctaca acgatatcaa ttt		33
SEQ ID NO: 149	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 149		
ctcccgaag gctagctaca acgacctttg gca		33
SEQ ID NO: 150	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 150		
gtacatggag gctagctaca acgatcaaag ttc		33
SEQ ID NO: 151	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Dnzyme	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 151		
aactggtag gctagctaca acgaggcaga ata		33
SEQ ID NO: 152	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = Scrambled dnzyme control	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 152		
tgtgatgtag gctagctaca acgacagatg gaa		33
SEQ ID NO: 153	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = GATA3 forward primer	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 153		
gaaccggccc cttatcaag		19

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SEQ ID NO: 154	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = GATA3 reverse primer	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 154		
acagttcgcg caggatgtc		19
SEQ ID NO: 155	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = GAPDH forward primer	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 155		
aggtcggtgt gaacggattt g		21
SEQ ID NO: 156	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = GAPDH reverse primer	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 156		
tgtagaccat gtagttgagg tca		23
SEQ ID NO: 157	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = IL-6 forward primer	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 157		
tagtccttcc taccccaatt tcc		23
SEQ ID NO: 158	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = IL-6 reverse primer	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 158		
ttggtcctta gccactcctt c		21
SEQ ID NO: 159	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = IL-10 forward primer	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 159		
gctcttactg actggcatga g		21
SEQ ID NO: 160	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = IL-10 reverse primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 160		
cgcagctcta ggagcatgtg		20

The invention claimed is:

1. A method of reducing insulin resistance in an individual in need thereof by inhibiting GATA3 expression, wherein the GATA3 expression is inhibited by administering to the subject an effective amount of a dnzyme having an oligonucleotide sequence selected from any one of SEQ ID NOs: 1-151. 5
2. The method of claim 1, wherein the dnzyme is hgd40 having an oligonucleotide sequence of SEQ ID NO:1.
3. The method of claim 1, wherein the individual is a mammal. 10
4. The method of claim 3, wherein the mammal is a human.
5. The method of claim 4, wherein the human is suffering from insulin resistance or Type II diabetes. 15
6. The method of claim 1, wherein the administering of the GATA3 inhibitor to the subject comprises increasing expression of IL-10 gene and decreasing expression of IL-6, PAI-I, resistin, and glucagon genes in the subject.
7. A method of reducing impaired adipogenesis in a subject in need thereof by inhibiting GATA3 expression, wherein the GATA3 expression is inhibited by administering to the subject an effective amount of a dnzyme having an oligonucleotide sequence selected from any one of SEQ ID NOs: 1-151. 20 25
8. The method of claim 7, wherein the dnzyme is hgd40 having an oligonucleotide sequence of SEQ ID NO:1.
9. The method of claim 7, wherein the patient is a mammal.
10. The method of claim 9, wherein the mammal is a human. 30

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